Retail Sales Analysis & Business Insights & Findings Between 2010 and 2011.

1. Project Overview

This project uses Microsoft Power BI to analyse an Online Retail dataset. The purpose is to understand sales trends, customer purchasing behaviour, product performance and geographical revenue distribution.

2. Business Questions and Findings

These are the five main business questions we used to analyse the dataset.

Question 1: Which year generated the highest revenue?

Finding: 2011

The analysis shows that 2011 generated higher revenue than 2010, indicating stronger revenue performance during the later reporting period. However, the comparison should consider that the dataset covers only part of 2011

Question 2: Which month recorded the highest revenue?

Finding: November

November achieved the highest monthly revenue, suggesting increased customer purchasing activity during this period.

Question 3: Which country generated the highest revenue?

Finding: United Kingdom

The United Kingdom was the leading contributor to revenue among the countries included in the analysis, highlighting its importance to overall business sales.

Question 4: Which product description generated the highest revenue?

Finding: DOTCOM POSTAGE

DOTCOM POSTAGE generated the highest revenue among the product descriptions analysed. It is important to note that this is a postage-related charge rather than a physical retail product.

Question 5: Which customer generated the highest revenue?

Finding: Customer ID 14646

Customer ID 14646 generated the highest revenue among the customers analysed, making this customer a significant contributor to overall sales.

3.Key Business Insights

The analysis provides five important insights:

1. Yearly performance: Revenue was higher in 2011 than in 2010, subject to the partial-year reporting period.

2. Monthly performance: November was the strongest month for revenue.
3. Geographical performance: The United Kingdom contributed the highest revenue.
4. Product performance: DOTCOM POSTAGE was the highest-revenue description, although it represents a postage charge.
5. Customer performance: Customer ID 14646 was the highest revenue customer.

4. Tools and Technologies Used

- Microsoft Power BI
- Power Query – Data cleaning and transformation
- DAX – Calculations and measures
- Data visualisation
- Business intelligence and data analysis

5. Business Value

This project helps identify revenue patterns, leading markets, important customers and product-related performance. These findings can support better sales planning, customer retention strategies and informed business decision-making.

6. Dashboard components

Your Power BI dashboard includes:

Visual	Purpose
Total Revenue card	Shows overall revenue
Total Orders card	Displays total orders
Total Customers card	Shows customer count
Units Sold card	Displays units sold
Revenue by Month	Identifies monthly sales trends
Revenue by Country	Compares geographical performance
Revenue by Year	Compares yearly performance
Revenue by Description	Identifies leading product descriptions
Top 10 Customers	Highlights important customers
Invoice Date slicer	Filters by date
Country slicer	Filters by country

7. Skills Demonstrated

- Data cleaning and preparation
- Data analysis
- DAX calculations
- Power BI dashboard development
- Data visualisation

- Business insight generation
- Analytical thinking and interpretation